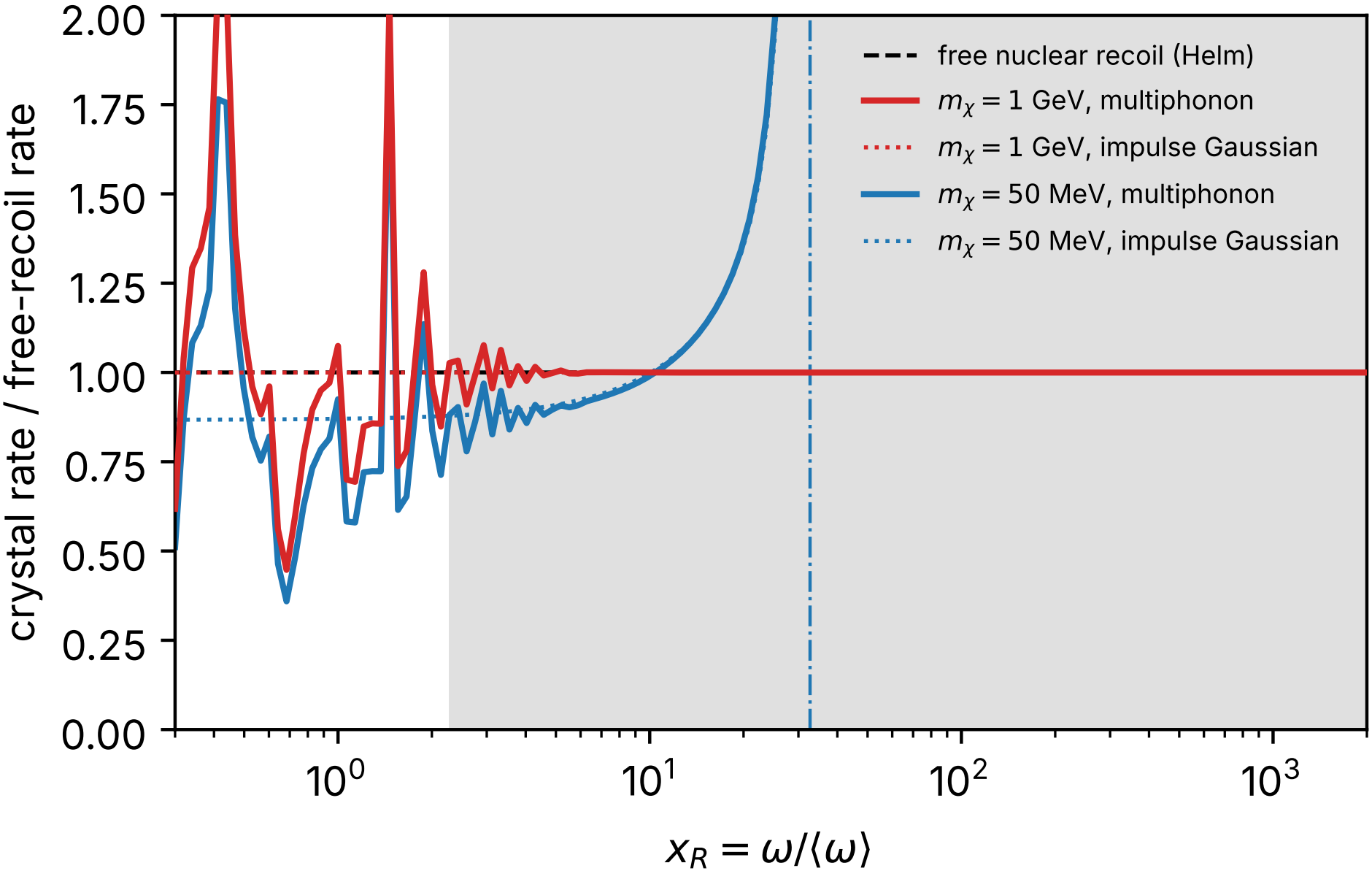
free nuclear recoil (Helm)
$m_\chi = 1$ GeV, multiphonon
$m_\chi = 1$ GeV, impulse Gaussian
$m_\chi = 50$ MeV, multiphonon
$m_\chi = 50$ MeV, impulse Gaussian
crystal rate / free-recoil rate
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
10^0
10^1
10^2
10^3
$x_R = \omega/\langle\omega\rangle$